SNS College of Technology

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23TSP201-GEN AI PROJECT

Topic -Lecture to Notes: A GenAI Learning Assistant

TEAM

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INNOVATION TECHNOLOGIES

- ROBOTICS & AUTOMATION
- AR/VR/META-VERSE/GAMING & DIGITAl TWIN
- DATA SCIENCE /AI/ ML
- INTERNET OF THINGS
- **COMMUNICATION AND GROWTH TECH**
- ADDITIVE MANUFACTURING (3D Printing)
- LOW CODE DEVELOPMENT

INDUSTRY VERTICALS

- **SMART CITY/MANUFACTURING**
- HEALTH CARE
- AGRICULTURE & FOOD TECHNOLOGY
- AUTOMOBILES
- AEROSPACE AND DEFENCE
- RETAIL, REAL ESTATE, ENTERTAINMENT & FINANCE
- POWER/ENERGY

EMPATHY

- students feel overwhelmed by hundreds of pages of lecture notes, slides, and textbooks before exam
- Time pressure: Many spend nights rewriting notes, but still fail to cover everything.
- Focus issues: Students don’t know which points are important vs. filler in their notes.

EMPATHY

- Unequal access: Some students have neat, detailed notes, while others rely on incomplete or messy versions.
- Stress & anxiety rise during exams because notes are disorganized and revision takes too long.
- Teachers feel burdened creating revision guides or summaries for students, which eats into their prep time..

EMPATHY

- Different learning styles: Some prefer text summaries, others flashcards, others quizzes — one-size-fits-all doesn't work.
- Group studies are inefficient because each student brings different note quality → confusion & wasted time.
- Digital-native students expect quick AI help, but current summarizers are too generic and don't adapt to their syllabus.
- Parents/mentors notice students spend more time rewriting notes than actually learning, which reduces real understanding.

DEFINE

- Students find it challenging to convert lectures into organized and concise notes for later study.
- College students attending lectures who need clear, structured, and easy-to-understand notes.
- Teachers who want to ensure students understand the lecture content effectively.
- An AI system that automatically converts lecture audio/video into summarized and accurate text notes.

DEFINE

- Manual note-taking divides attention between listening and writing, leading to poor comprehension
- To obtain quick, high-quality, and well-formatted notes after lectures without extra effort.
- In classrooms, online lectures, or recorded sessions where note generation is required.
- Automatically generated, editable, and shareable notes that enhance learning efficiency.

DEFINE

- The system should support speech-to-text, summarization, and content structuring features
- It must be accurate, fast, user-friendly, and accessible on multiple devices.
- Notes should be clear, concise, grammatically correct, and highlight important concepts.

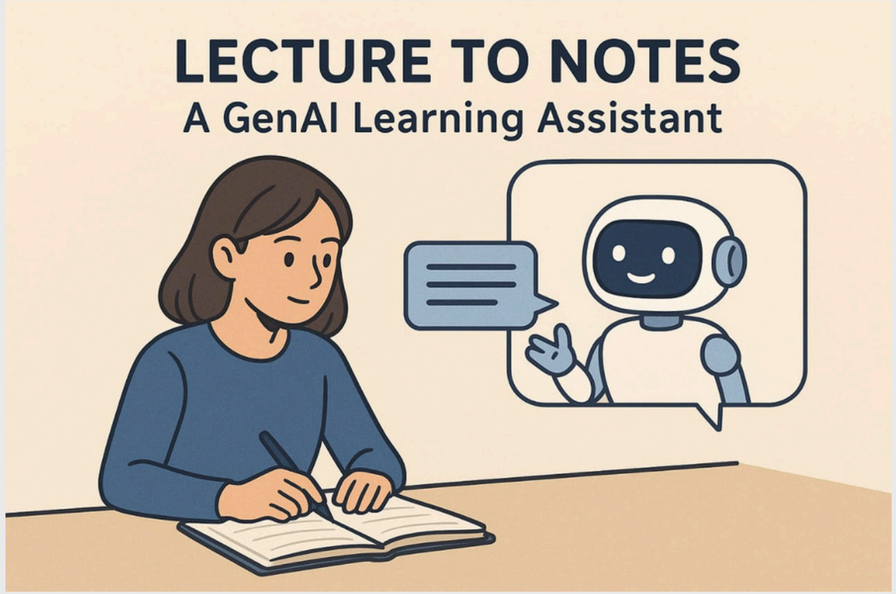
DEFINE

- Use of GenAI, Natural Language Processing (NLP), and Speech Recognition technologies.
- The solution should serve online adapting to different accents and subjects.
- Students and educators need a smart GenAI assistant that transforms spoken lectures into well-organized, concise, and accessible digital notes to improve learning productivity

IDEATE

- Discussed with students and teachers to find creative ways to make lecture-to-note conversion faster and smarter.

- Develop a GenAI assistant that listens to lectures and automatically creates summarized, structured, and clear notes.



IDEATE

- Speech-to-Text: Convert lecture audio to text in real time.
- Summarization: Extract key points and concepts using NLP.
- Multi-Format Notes: Provide text, bullets, or mind map formats.
- Smart Highlighting: Auto-tag important terms and examples.

IDEATE

- Personalization: Adapt notes to student learning style.
- Visual Aids: Generate diagrams or keyword visuals.
- LMS Integration: Connect with platforms like Google Classroom.